Benford-Weighted Digit Sums and Second-Order Asymptotics for Exponential Sequences

Radhe Dhoundiyal
Independent Researcher

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Abstract

Let $b \geq 2$ be an integer base, and let M_L denote the L-digit integer significand induced by the logarithmic, or Benford, mantissa measure. Starting from the classical prefix law

$$\mathbb{P}(M_L = m) = \log_b\left(\frac{m+1}{m}\right),$$

we derive an exact finite evaluation for the Benford-weighted base-b digit sum $\mathbb{E}[s_b(M_L)]$. This yields an exact finite formula for the cumulative signed deviation of significant-digit expectations from the uniform digit mean and, in the limit,

$$c_b = \frac{b}{2} - \frac{b-1}{2}\log_b(2\pi).$$

We also obtain a two-term finite-length expansion with an explicit Stirling-remainder bound.

We then transfer this constant to deterministic exponential sequences. For multiplicatively independent integers $a, b \geq 2$, a finite discrepancy estimate shows that the mean sum of the first J significant digits of a^n equals

$$J\frac{b-1}{2} - c_b$$

up to explicit errors. Combining this with classical quantitative discrepancy bounds for $\{n \log_b a\}$ gives, for a logarithmically growing leading window J_N,

$$\mathcal{S}_{N,J_N}(a;b) = J_N\frac{b-1}{2} - c_b + o(1).$$

Thus the same cumulative Benford bias appears as a bounded second-order correction in a deterministic exponential sequence.

1 Introduction and main results

The significant digits of exponential sequences provide a classical meeting point of uniform distribution theory, Benford's law, and digital number theory. If $b \geq 2$ is an integer base and Θ is uniformly distributed on $[0, 1)$, then the random mantissa

$$X = b^{\Theta} \in [1, b)$$

induces the logarithmic, or Benford, distribution on significant digits. Equivalently, questions about the significant digits of deterministic powers a^n are naturally related to the rotation

$$\{n \log_b a\},$$

whenever a and b are multiplicatively independent.

The distribution of a fixed significant prefix in the logarithmic mantissa model is classical. In particular, if M_L denotes the integer formed by the first L significant base-b digits of b^Θ, then

$$\mathbb{P}(M_L = m) = \log_b\left(\frac{m+1}{m}\right), \qquad b^{L-1} \le m \le b^L - 1. \tag{1}$$

The corresponding marginal laws of individual significant digits, their relation to uniform distribution modulo one, and the convergence of deeper significant digits toward the uniform digit distribution are also well established; see, among others, [18, 4, 8, 13, 14, 5, 6]. Closed Gamma-function expressions for the individual higher-digit marginals were obtained by Balado and Silvestre [2, 3]. We do not claim novelty for these fixed-position distribution laws.

The purpose of this paper is instead to study an aggregate quantity that retains the systematic bias of the early significant positions.

Let $D_j^{(b)}$ denote the j-th significant base-b digit under the logarithmic mantissa model, and set

$$\mu_b = \frac{b-1}{2},$$

the mean of a uniformly distributed digit in $\{0, \dots, b-1\}$. For $L \ge 1$, define the finite cumulative signed significant-digit bias by

$$c_{b,L} = \sum_{j=1}^{L} \left(\mu_b - \mathbb{E}[D_j^{(b)}]\right). \tag{2}$$

The adjective "signed" is important: the quantity need not be positive for every base.

Our first main result gives an exact finite evaluation of the corresponding Benford-weighted digit sum. Let $s_b(m)$ denote the ordinary sum of the base-b digits of the integer m.

Theorem 1.1 (Exact Benford-weighted digit sum)**.** *For every integer $b \ge 2$ and every $L \ge 1$,*

$$\boxed{\mathbb{E}[s_b(M_L)] = b^L - b + L(b-1) - \log_b\left(\frac{(b^L-1)!}{\left((b^{L-1}-1)!\right)^b}\right).} \tag{3}$$

Equivalently,

$$\boxed{c_{b,L} = \frac{\ln\Gamma(b^L) - b\ln\Gamma(b^{L-1})}{\ln b} - (b^L - b) - \frac{L(b-1)}{2}.} \tag{4}$$

The proof uses the classical prefix law (1) together with the elementary base-b recursion

$$s_b(bq + r) = s_b(q) + r, \qquad 0 \le r \le b-1.$$

Although the probability law in (1) is classical, we have not found the weighted evaluation (3), or the equivalent finite cumulative identity (4), in the literature surveyed.

Passing to the limit produces a simple closed constant.

Theorem 1.2 (Cumulative significant-digit bias)**.** *For every integer base $b \ge 2$, the series*

$$c_b = \sum_{j=1}^{\infty} \left(\frac{b-1}{2} - \mathbb{E}[D_j^{(b)}]\right)$$

converges absolutely and satisfies

$$\boxed{c_b = \frac{b}{2} - \frac{b-1}{2}\log_b(2\pi).} \tag{5}$$

For example, in base 10,

$$c_{10} = 5 - \frac{9}{2}\log_{10}(2\pi) = 1.408190592388482\ldots. \tag{6}$$

Thus the cumulative effect of the biased early significant positions is governed by the same 2π constant that appears in Stirling's formula.

The finite identity also gives precise control of the approach to the limit.

Theorem 1.3 (Finite-length expansion)**.** *For every $b \geq 2$ and $L \geq 1$,*

$$c_b - c_{b,L} = \frac{b^2-1}{12\ln b}\, b^{-L} - \frac{b^4-1}{360\ln b}\, b^{-3L} + \varepsilon_{b,L}, \tag{7}$$

where

$$\boxed{|\varepsilon_{b,L}| \leq \frac{b^6+1}{1260\ln b}\, b^{-5L}.} \tag{8}$$

The individual significant-digit distributions admit corresponding deep-position expansions. These are useful below, but are not the principal novelty of the paper. Expanding the known exact higher-significant-digit law yields, for $j \geq 2$ and $r \in \{0,\ldots,b-1\}$,

$$\mathbb{P}(D_j^{(b)} = r) = \frac{1}{b} + \frac{(b-1)(b-1-2r)}{2b\ln b}\, b^{-j} + O_b(b^{-2j}), \tag{9}$$

and consequently

$$\mathbb{E}[D_j^{(b)}] = \frac{b-1}{2} - \frac{(b-1)^2(b+1)}{12\ln b}\, b^{-j} + O_b(b^{-2j}). \tag{10}$$

These formulas should be viewed as quantitative consequences of the known higher-digit distribution rather than as independent distribution theorems.

1.1 From the logarithmic model to exponential sequences

The second aim of the paper is to determine whether the constant c_b, obtained in the logarithmic mantissa model, persists in an actual deterministic exponential sequence.

Let $a, b \geq 2$ be multiplicatively independent integers and write

$$\alpha = \log_b a.$$

Then α is irrational. For $n \geq 1$, let

$$L_n = \lfloor n\alpha \rfloor + 1$$

be the number of base-b digits of a^n, and let $D_j^{(b)}(a^n)$ denote its j-th significant digit.

For $N, J \geq 1$, define the leading-window digit-sum average

$$\mathcal{S}_{N,J}(a;b) = \frac{1}{N} \sum_{\substack{1 \leq n \leq N \\ L_n \geq J}} \sum_{j=1}^{J} D_j^{(b)}(a^n). \tag{11}$$

The normalization is by N; powers with fewer than J digits contribute nothing to the sum.

Let $D_N(\alpha)$ denote the ordinary interval discrepancy of

$$\{\alpha\}, \{2\alpha\}, \ldots, \{N\alpha\}.$$

That is,

$$D_N(\alpha) = \sup_{0\le u<v\le 1} \left| \frac{1}{N}\#\{1 \le n \le N : \{n\alpha\} \in [u,v)\} - (v-u) \right|.$$

The digit cylinders associated with the first J significant positions have exponentially increasing combinatorial complexity, but ordinary discrepancy still gives an explicit finite transfer estimate.

Theorem 1.4 (Finite deterministic transfer)**.** *Let $a, b \ge 2$ be multiplicatively independent. Let*

$$e_J(N) = \#\{1 \le n \le N : L_n < J\}.$$

Then

$$\boxed{\left| \mathcal{S}_{N,J}(a;b) - \left(J\frac{b-1}{2} - c_{b,J} \right) \right| \le \frac{b-1}{2}\, b^J D_N(\alpha) + \frac{(b-1)J\, e_J(N)}{N}.} \tag{12}$$

Since

$$e_J(N) = O_{a,b}(J),$$

Theorem 1.4 and Theorem 1.3 imply the following second-order asymptotic.

Theorem 1.5 (Second-order leading-window asymptotic)**.** *Let $J = J(N)$ satisfy*

$$J(N) \longrightarrow \infty, \qquad b^{J(N)} D_N(\log_b a) \longrightarrow 0, \qquad \frac{J(N)^2}{N} \longrightarrow 0.$$

Then

$$\boxed{\mathcal{S}_{N,J(N)}(a;b) = J(N)\frac{b-1}{2} - c_b + o(1).} \tag{13}$$

More quantitatively,

$$\mathcal{S}_{N,J}(a;b) = J\frac{b-1}{2} - c_b + O_b(b^{-J}) + O_b\big(b^J D_N(\log_b a)\big) + O_{a,b}\left(\frac{J^2}{N}\right). \tag{14}$$

The point of (13) is not the first-order term $J(b-1)/2$: first-order digit and block statistics in growing regions of exponential sequences have been studied previously by discrepancy and exponential-sum methods, notably by Fuchs [12] and the literature cited there. The distinction here is that the accumulated nonuniformity of the significant positions is retained on the absolute $O(1)$ scale, and its limiting contribution is exactly the same constant c_b that arises in the logarithmic mantissa model.

Classical quantitative discrepancy estimates for $\alpha = \log_b a$, based ultimately on Diophantine estimates for linear forms in logarithms, yield a power saving for every fixed multiplicatively independent pair (a, b); see [12, 16, 17, 1, 10]. Consequently, one obtains logarithmically growing windows as a direct corollary.

Corollary 1.6 (Logarithmically growing deterministic window)**.** *Fix multiplicatively independent integers $a, b \ge 2$. Suppose*

$$D_N(\log_b a) \ll_{a,b} N^{-\delta_{a,b}}$$

for some $\delta_{a,b} > 0$, as supplied by the classical quantitative discrepancy theory cited above. For any

$$0 < \kappa < \delta_{a,b}, \qquad J_N = \lfloor \kappa \log_b N \rfloor ,$$

one has

$$\boxed{\mathcal{S}_{N,J_N}(a;b) = J_N \frac{b-1}{2} - c_b + o(1).} \tag{15}$$

Thus the constant c_b has two interpretations. It is first the total signed positional bias of the significant digits under the logarithmic mantissa measure, and it then reappears as a deterministic boundary correction for a growing leading block of the powers a^n.

1.2 Relation to prior work

The logarithmic law for leading significant digits goes back to Newcomb and Benford [18, 4]. The relation between leading-digit phenomena and uniform distribution modulo one was developed systematically by Diaconis [8]. Hill's work [13, 14] and the later treatment of Berger and Hill [5, 6] provide a modern framework for significant-digit distributions and Benford measures. Regularity of deeper significant digits has also been studied in considerably greater generality [15].

Particularly relevant here is the work of Balado and Silvestre [2, 3], who give an exact Gamma-function representation for the marginal distribution of the j-th significant digit of a Benford variable. That formula is prior work and is not claimed here. The position-wise expansion (9) may be obtained by expanding this known exact distribution.

The ordinary sum-of-digits function has an extensive literature, including the classical work of Trollope and Delange and later Mellin-transform approaches [19, 7, 11]. Those results primarily concern counting or summation over integers with uniform arithmetic weights. The finite expectation considered in Theorem 1.1 instead weights the L-digit integer significands according to the logarithmic masses

$$\log_b \frac{m+1}{m}.$$

To the best of our knowledge, the resulting exact factorial identity and the closed evaluation (5) have not previously been recorded.

For deterministic exponential sequences, Fuchs [12] established quantitative first-order results for digital blocks in growing regions using discrepancy and exponential sums, building on earlier work in discrepancy and Diophantine approximation. We therefore do not claim novelty for the use of discrepancy, for first-order uniform digit frequencies, or for the existence of logarithmically growing controlled windows. Our second-order statement retains the bounded contribution left after subtracting the uniform first-order digit mean. We have not found the explicit constant $-c_b$, or an equivalent evaluation, in the earlier exponential-sequence literature surveyed.

Accordingly, the principal contribution of the present paper is the chain

$$\boxed{\begin{array}{c} \text{classical Benford prefix law} \\ \Downarrow \\ \text{exact Benford-weighted finite digit-sum identity} \\ \Downarrow \\ c_b = \dfrac{b}{2} - \dfrac{b-1}{2}\log_b(2\pi) \\ \Downarrow \\ \text{the same } c_b \text{ as a deterministic second-order correction.} \end{array}}$$

1.3 Scope and organization

The deterministic results in this paper concern a growing block of *leading significant digits*. They do not establish an asymptotic law for the complete base-b digit sum of a^n, whose number of digits is of order n. In particular, no independence, mixing, central limit theorem, or variance law for the complete digit sequence is assumed or proved here.

Section 2 recalls the logarithmic significand model and the classical significant-digit laws needed below. Section 3 proves the exact finite Benford-weighted digit-sum identity and evaluates the cumulative constant c_b. Section 4 develops the finite-position and finite-length asymptotics. Section 5 transfers the resulting constant to deterministic exponential sequences by discrepancy estimates. Section 6 records the independent computational checks used to audit the formulas, Section 7 discusses the limitations of the method and several open problems, and Section 8 concludes the paper.

2 Benford prefixes and significant digits

This section fixes notation and records the classical significant-digit facts used later. None of the distribution laws in this section are claimed as new. The finite prefix law is standard Benford theory, and the Gamma-function form of the higher significant-digit marginals is due to Balado and Silvestre [2, 3].

Throughout, $b \geq 2$ is an integer base and

$$\Theta \sim \mathrm{Unif}[0,1).$$

We write

$$X = b^{\Theta} \in [1, b).$$

Equivalently, X has density

$$\frac{1}{x \ln b}, \qquad 1 \leq x < b,$$

with respect to Lebesgue measure.

For $j \geq 1$, let $D_j^{(b)}$ denote the j-th significant base-b digit of X. Thus

$$D_1^{(b)} \in \{1, \ldots, b-1\},$$

whereas

$$D_j^{(b)} \in \{0, \ldots, b-1\}, \qquad j \geq 2.$$

It will also be convenient to use the ordinary base-b digit-sum function

$$s_b(m),$$

defined for nonnegative integers m.

2.1 Finite integer significands

For $L \geq 1$, define the L-digit integer significand

$$M_L = \left\lfloor b^{L-1+\Theta} \right\rfloor. \tag{16}$$

Then

$$b^{L-1} \leq M_L \leq b^L - 1,$$

and the base-b digits of M_L are precisely

$$D_1^{(b)}, D_2^{(b)}, \ldots, D_L^{(b)}.$$

Consequently,

$$s_b(M_L) = \sum_{j=1}^{L} D_j^{(b)}. \tag{17}$$

The distribution of M_L is the standard Benford distribution on L-digit integer significands.

Proposition 2.1 (Classical Benford prefix law)**.** *For*

$$b^{L-1} \le m \le b^L - 1,$$

one has

$$\mathbb{P}(M_L = m) = \log_b\left(\frac{m+1}{m}\right). \tag{18}$$

Proof. By definition,

$$M_L = m$$

if and only if

$$m \le b^{L-1+\Theta} < m+1.$$

Taking logarithms gives

$$\log_b\left(\frac{m}{b^{L-1}}\right) \le \Theta < \log_b\left(\frac{m+1}{b^{L-1}}\right).$$

Since Θ is uniform on $[0,1)$, the probability is the length of this interval:

$$\log_b(m+1) - \log_b m = \log_b\left(\frac{m+1}{m}\right).$$

□

The proof is included only to establish notation and normalization. Equation (18) is classical; see, for example, [8, 13, 5, 6].

For $L = 1$, Proposition 2.1 yields the usual first-digit law

$$\mathbb{P}(D_1^{(b)} = d) = \log_b\left(1 + \frac{1}{d}\right), \qquad d = 1, \ldots, b-1. \tag{19}$$

A direct finite summation also gives

$$\mathbb{E}[D_1^{(b)}] = (b-1) - \log_b((b-1)!). \tag{20}$$

We record (20) for later comparison; it is an elementary consequence of the classical leading-digit law.

2.2 Individual higher significant digits

Let $j \geq 2$ and

$$r \in \{0, \ldots, b-1\}.$$

Put

$$M = b^{j-2}. \tag{21}$$

A j-digit prefix whose final digit is r has the form

$$bq + r,$$

where

$$M \leq q \leq bM - 1.$$

Summing the classical prefix probabilities therefore gives the j-th digit marginal.

Proposition 2.2 (Classical higher-digit marginal)**.** *For $j \geq 2$,*

$$\boxed{\mathbb{P}(D_j^{(b)} = r) = \sum_{q=M}^{bM-1} \log_b \left(\frac{bq + r + 1}{bq + r} \right).} \tag{22}$$

Proof. The events

$$\{M_j = bq + r\}, \qquad M \leq q \leq bM - 1,$$

are disjoint and exhaust the possible j-digit prefixes having final digit r. Applying Proposition 2.1 with $L = j$ and summing the corresponding probabilities gives (22). □

An equivalent closed form is obtained by converting the finite product implicit in (22) to Gamma functions. This representation is prior work of Balado and Silvestre [2, 3].

Proposition 2.3 (Balado–Silvestre Gamma representation)**.** *For $j \geq 2$, $M = b^{j-2}$, and $r \in \{0, \ldots, b-1\}$,*

$$\boxed{\mathbb{P}(D_j^{(b)} = r) = \frac{1}{\ln b} \ln \left[\frac{\Gamma\left(bM + \frac{r+1}{b}\right) \Gamma\left(M + \frac{r}{b}\right)}{\Gamma\left(bM + \frac{r}{b}\right) \Gamma\left(M + \frac{r+1}{b}\right)} \right].} \tag{23}$$

Proof. From (22),

$$\mathbb{P}(D_j^{(b)} = r) = \frac{1}{\ln b} \ln \prod_{q=M}^{bM-1} \frac{q + (r+1)/b}{q + r/b}.$$

Using

$$\prod_{q=M}^{bM-1} (q + \lambda) = \frac{\Gamma(bM + \lambda)}{\Gamma(M + \lambda)}$$

gives (23). □

Remark (Attribution). The derivation above is included for completeness and to fix the present notation. The Gamma representation (23) is not a contribution of this paper. Balado and Silvestre gave this form in their 2021 conference paper and in their later general treatment of number-representation elements [2, 3].

The probabilities in (22) satisfy

$$\sum_{r=0}^{b-1} \mathbb{P}(D_j^{(b)} = r) = 1.$$

Moreover,

$$\mathbb{P}(D_j^{(b)} = r) \longrightarrow \frac{1}{b} \qquad (j \to \infty),$$

a classical manifestation of the asymptotic regularity of deeper significant digits [13, 15, 3]. A quantitative first correction to this limit will be recorded in Section 4, where it is used as supporting information rather than as a principal novelty claim.

2.3 From position-wise laws to a finite digit sum

By (17) and linearity of expectation,

$$\mathbb{E}[s_b(M_L)] = \sum_{j=1}^{L} \mathbb{E}[D_j^{(b)}]. \tag{24}$$

Thus one could, in principle, study the cumulative effect of the nonuniform significant positions by summing the individual marginal expectations.

For the main results, however, it is more effective to work directly with the finite random integer M_L. Combining Proposition 2.1 with (17) gives

$$\mathbb{E}[s_b(M_L)] = \sum_{m=b^{L-1}}^{b^L-1} s_b(m) \log_b\left(\frac{m+1}{m}\right). \tag{25}$$

Equation (25) is the starting point of the exact finite evaluation in the next section. The probability weights are classical; the issue is whether the logarithmically weighted digital sum itself admits a useful closed form. Section 3 shows that it does.

3 Exact Benford-weighted digit sums and cumulative bias

We now evaluate the logarithmically weighted digit sum introduced in (25). The input distribution is the classical Benford prefix law; the new ingredient is that the weighted sum collapses to a finite factorial expression.

For convenience, write

$$E_{b,L} = \mathbb{E}[s_b(M_L)]. \tag{26}$$

By Proposition 2.1,

$$E_{b,L} = \sum_{m=b^{L-1}}^{b^L-1} s_b(m) \log_b\left(\frac{m+1}{m}\right). \tag{27}$$

The elementary identity

$$s_b(bq + r) = s_b(q) + r, \qquad 0 \le r \le b-1, \tag{28}$$

allows the sum to be reduced recursively.

3.1 A finite block identity

We first isolate the contribution of the final base-b digit.

Lemma 3.1. *Let $L \geq 2$, and put*

$$A = b^{L-2}, \qquad B = b^{L-1} = bA.$$

Then

$$E_{b,L} = E_{b,L-1} + R_{b,L}, \tag{29}$$

where

$$\boxed{R_{b,L} = (b-1)(B+1) - \log_b \left(\frac{(bB-1)!\,(A-1)!^b}{(B-1)!^{\,b+1}} \right).} \tag{30}$$

Proof. Every integer

$$m \in [B, bB-1]$$

can be written uniquely as

$$m = bq + r,$$

with

$$A \leq q \leq B-1, \qquad 0 \leq r \leq b-1.$$

Using (28) in (27) gives

$$\begin{aligned} E_{b,L} &= \sum_{q=A}^{B-1} \sum_{r=0}^{b-1} \big(s_b(q) + r\big) \log_b \left(\frac{bq+r+1}{bq+r} \right) \\ &= \sum_{q=A}^{B-1} s_b(q) \sum_{r=0}^{b-1} \log_b \left(\frac{bq+r+1}{bq+r} \right) + R_{b,L}, \end{aligned} \tag{31}$$

where

$$R_{b,L} = \sum_{q=A}^{B-1} \sum_{r=0}^{b-1} r \log_b \left(\frac{bq+r+1}{bq+r} \right). \tag{32}$$

The inner logarithmic sum in the first term telescopes:

$$\sum_{r=0}^{b-1} \log_b \left(\frac{bq+r+1}{bq+r} \right) = \log_b \left(\frac{q+1}{q} \right).$$

Hence the first term in (31) is precisely

$$E_{b,L-1}.$$

It remains to evaluate $R_{b,L}$. For fixed q, discrete summation by parts gives

$$\begin{aligned} &\sum_{r=0}^{b-1} r \log_b \left(\frac{bq+r+1}{bq+r} \right) \\ &\qquad = \log_b \left(\frac{(bq+b)^{b-1}}{\prod_{r=1}^{b-1}(bq+r)} \right). \end{aligned} \tag{33}$$

Multiplying (33) over $A \le q \le B-1$ yields

$$R_{b,L} = \log_b \left[\frac{\prod_{q=A}^{B-1} (b(q+1))^{b-1}}{\prod_{q=A}^{B-1} \prod_{r=1}^{b-1} (bq+r)} \right]. \tag{34}$$

The numerator is

$$b^{(b-1)(B-A)} \left(\frac{B!}{A!} \right)^{b-1}. \tag{35}$$

The denominator consists of all integers from $B+1$ to $bB-1$ that are not divisible by b. Therefore

$$\begin{aligned} \prod_{q=A}^{B-1} \prod_{r=1}^{b-1} (bq+r) &= \frac{(bB-1)!/B!}{\prod_{k=A+1}^{B-1} bk} \\ &= \frac{(bB-1)!\, A!}{B!\, b^{B-A-1} (B-1)!}. \end{aligned} \tag{36}$$

Substituting (35) and (36) into (34), and using

$$B = bA,$$

one obtains

$$\frac{\prod_{q=A}^{B-1} (b(q+1))^{b-1}}{\prod_{q=A}^{B-1} \prod_{r=1}^{b-1} (bq+r)} = b^{(b-1)(B+1)} \frac{(B-1)!^{\,b+1}}{(A-1)!^b\, (bB-1)!}. \tag{37}$$

Taking $\log_b$ gives (30). □

3.2 Exact finite Benford-weighted digit sum

We can now prove the first main theorem stated in the introduction.

Proof of Theorem 1.1. We argue by induction on L.

For $L = 1$,

$$M_1 = D_1^{(b)},$$

and hence

$$\begin{aligned} E_{b,1} &= \sum_{d=1}^{b-1} d \log_b \left(\frac{d+1}{d} \right) \\ &= (b-1) - \log_b((b-1)!). \end{aligned} \tag{38}$$

This agrees with (3), since $0! = 1$.

Now suppose $L \geq 2$ and that

$$E_{b,L-1} = b^{L-1} - b + (L-1)(b-1) - \log_b \left(\frac{(b^{L-1}-1)!}{((b^{L-2}-1)!)^b} \right). \tag{39}$$

Again put

$$A = b^{L-2}, \qquad B = b^{L-1}.$$

By Lemma 3.1,

$$\begin{aligned} E_{b,L} = E_{b,L-1} &+ (b-1)(B+1) \\ &- \log_b \left(\frac{(bB-1)!\,(A-1)!^b}{(B-1)!^{\,b+1}} \right). \end{aligned} \tag{40}$$

The elementary terms satisfy

$$B - b + (L-1)(b-1) + (b-1)(B+1) = bB - b + L(b-1).$$

Since $bB = b^L$, this is

$$b^L - b + L(b-1).$$

The logarithmic terms combine as

$$\begin{aligned} &\frac{(B-1)!}{(A-1)!^b} \cdot \frac{(bB-1)!\,(A-1)!^b}{(B-1)!^{\,b+1}} \\ &\qquad = \frac{(bB-1)!}{(B-1)!^b} = \frac{(b^L-1)!}{((b^{L-1}-1)!)^b}. \end{aligned} \tag{41}$$

Therefore

$$\boxed{E_{b,L} = b^L - b + L(b-1) - \log_b \left(\frac{(b^L-1)!}{((b^{L-1}-1)!)^b} \right),}$$

which completes the induction. □

Remark. The probability measure used in Theorem 1.1 is not new:

$$\mathbb{P}(M_L = m) = \log_b \left(\frac{m+1}{m} \right)$$

is the classical Benford law for the L-digit integer significand. The content of the theorem is the exact evaluation obtained after inserting the digital statistic $s_b(m)$ into this logarithmically weighted measure.

3.3 Finite cumulative significant-digit bias

Recall that

$$\mu_b = \frac{b-1}{2}$$

and

$$c_{b,L} = \sum_{j=1}^{L} \left(\mu_b - \mathbb{E}[D_j^{(b)}] \right).$$

Since

$$s_b(M_L) = \sum_{j=1}^{L} D_j^{(b)},$$

linearity of expectation gives

$$c_{b,L} = L\frac{b-1}{2} - E_{b,L}. \tag{42}$$

Substituting Theorem 1.1 into (42) gives

$$c_{b,L} = \log_b\left(\frac{(b^L-1)!}{((b^{L-1}-1)!)^b}\right) - (b^L - b) - \frac{L(b-1)}{2}. \tag{43}$$

Using

$$n! = \Gamma(n+1)$$

yields the equivalent Gamma representation

$$\boxed{c_{b,L} = \frac{\ln\Gamma(b^L) - b\ln\Gamma(b^{L-1})}{\ln b} - (b^L - b) - \frac{L(b-1)}{2}.} \tag{44}$$

Thus the cumulative effect of the first L significant positions has an exact finite form even though the individual digit distributions are not uniform.

3.4 The limiting bias constant

We next evaluate the limit of (44).

Proof of Theorem 1.2. Set

$$x = b^{L-1},$$

so that

$$b^L = bx.$$

The classical Stirling expansion gives

$$\ln\Gamma(x) = \left(x - \frac{1}{2}\right)\ln x - x + \frac{1}{2}\ln(2\pi) + O(x^{-1}) \qquad (x \to \infty). \tag{45}$$

Applying (45) at x and bx gives

$$\begin{aligned}\ln\Gamma(bx) - b\ln\Gamma(x) = \left(bx - \frac{1}{2}\right)\ln b + \frac{b-1}{2}\ln x \\ - \frac{b-1}{2}\ln(2\pi) + O(x^{-1}).\end{aligned} \tag{46}$$

Since

$$\ln x = (L-1)\ln b,$$

division by $\ln b$ yields

$$\begin{aligned}\frac{\ln\Gamma(b^L) - b\ln\Gamma(b^{L-1})}{\ln b} = b^L - \frac{b}{2} + \frac{L(b-1)}{2} \\ - \frac{b-1}{2}\log_b(2\pi) + O_b(b^{-L}).\end{aligned} \tag{47}$$

Substituting this into (44), all terms depending linearly on b^L and L cancel. We obtain

$$c_{b,L} = \frac{b}{2} - \frac{b-1}{2}\log_b(2\pi) + O_b(b^{-L}). \tag{48}$$

Therefore

$$\boxed{c_b = \lim_{L\to\infty} c_{b,L} = \frac{b}{2} - \frac{b-1}{2}\log_b(2\pi).} \tag{49}$$

The absolute convergence of

$$\sum_{j\geq 1}\left(\frac{b-1}{2} - \mathbb{E}[D_j^{(b)}]\right)$$

follows from the quantitative higher-position estimate proved in Section 4. □

Remark (Signed nature of the bias). The term "bias" in this paper is signed. For example,

$$c_2 < 0, \qquad c_3 < 0,$$

whereas

$$c_{10} = 5 - \frac{9}{2}\log_{10}(2\pi) = 1.408190592388482\ldots.$$

Thus it would be misleading to call c_b a universal "digit deficit" across all bases.

3.5 Expected average of a finite significant prefix

Define

$$A_L^{(b)} = \frac{1}{L}\sum_{j=1}^{L} D_j^{(b)} = \frac{s_b(M_L)}{L}. \tag{50}$$

Then the definition of $c_{b,L}$ gives the exact identity

$$\boxed{\mathbb{E}[A_L^{(b)}] = \frac{b-1}{2} - \frac{c_{b,L}}{L}.} \tag{51}$$

Consequently,

$$\mathbb{E}[A_L^{(b)}] = \frac{b-1}{2} - \frac{c_b}{L} + O_b\left(\frac{b^{-L}}{L}\right), \tag{52}$$

with the sharper expansion and explicit remainder given in Section 4.

Equation (52) provides the first interpretation of c_b: it is the finite boundary correction left by the nonuniform early significant positions after the bulk digit mean has approached the uniform value $(b-1)/2$. In Section 5, we show that the same constant survives when the logarithmic mantissa measure is replaced by a quantitative orbit average along the deterministic sequence a^n.

4 Quantitative finite-position and finite-length asymptotics

The preceding section evaluated the cumulative significant-digit bias exactly. We now record two quantitative consequences.

First, the known exact distribution of an individual higher significant digit admits a simple first-order expansion about the uniform digit law. Second, the exact finite formula for $c_{b,L}$ yields a two-term asymptotic expansion with an explicit Stirling-remainder bound.

The first result is included primarily as quantitative background for the cumulative theory. It should not be interpreted as a new significant-digit distribution theorem: the exact marginal law and its Gamma representation are prior work, as discussed in Section 2.2.

4.1 Asymptotics of an individual significant digit

Let $j \geq 2$, and write

$$M = b^{j-2}.$$

For

$$r \in \{0, \ldots, b-1\},$$

Proposition 2.3 gives

$$\mathbb{P}(D_j^{(b)} = r) = \frac{1}{\ln b} \left[\Delta(bM) - \Delta(M)\right], \tag{53}$$

where

$$\Delta(z) = \ln \Gamma(z+u) - \ln \Gamma(z+v),$$

with

$$u = \frac{r+1}{b}, \qquad v = \frac{r}{b}.$$

We use the standard fixed-parameter Gamma-ratio expansion [9, §5.11(iii)]

$$\ln \Gamma(z+u) - \ln \Gamma(z+v) = (u-v)\ln z + \frac{(u-v)(u+v-1)}{2z} + O_{u,v}(z^{-2}) \tag{54}$$

as $z \to \infty$. Since b is fixed and r ranges over only finitely many values, the remainder may be taken uniformly in $r \in \{0, \ldots, b-1\}$, with a constant depending only on b.

Proposition 4.1 (Deep-position expansion)**.** *For fixed $b \geq 2$, uniformly in $r \in \{0, \ldots, b-1\}$,*

$$\boxed{\mathbb{P}(D_j^{(b)} = r) = \frac{1}{b} + \frac{(b-1)(b-1-2r)}{2b \ln b}\, b^{-j} + O_b(b^{-2j})} \tag{55}$$

as $j \to \infty$.

Proof. From

$$u - v = \frac{1}{b}$$

and

$$u + v - 1 = \frac{2r+1-b}{b},$$

equation (54) gives

$$\Delta(z) = \frac{1}{b}\ln z + \frac{2r+1-b}{2b^2 z} + O_b(z^{-2}). \tag{56}$$

Applying this at $z = bM$ and $z = M$, we obtain

$$\begin{aligned} \Delta(bM) - \Delta(M) = \frac{1}{b}\ln b \\ + \frac{2r+1-b}{2b^2}\left(\frac{1}{bM} - \frac{1}{M}\right) + O_b(M^{-2}). \end{aligned} \tag{57}$$

Since

$$\frac{1}{bM} - \frac{1}{M} = -\frac{b-1}{bM},$$

the correction term becomes

$$\frac{(b-1)(b-1-2r)}{2b^3 M}.$$

Using

$$M = b^{j-2}, \qquad \frac{1}{b^3 M} = \frac{1}{b}\, b^{-j},$$

and dividing by $\ln b$ in (53) yields (55). Finally,

$$M^{-2} = b^{-2j+4} = O_b(b^{-2j}),$$

which gives the stated remainder. □

Remark (Relation to prior work). The convergence

$$\mathbb{P}(D_j^{(b)} = r) \longrightarrow \frac{1}{b}$$

is classical, and the exact Gamma representation from which Proposition 4.1 is derived is due to Balado and Silvestre [2, 3]. Thus (55) is best viewed as an explicit first correction obtained by expanding a known exact formula.

4.2 Expectation of a deep significant digit

Summing the preceding expansion against the digit value gives the corresponding mean.

Corollary 4.2. *For fixed $b \geq 2$,*

$$\boxed{\mathbb{E}[D_j^{(b)}] = \frac{b-1}{2} - \frac{(b-1)^2(b+1)}{12 \ln b}\, b^{-j} + O_b(b^{-2j})} \tag{58}$$

as $j \to \infty$.

Proof. By definition,

$$\mathbb{E}[D_j^{(b)}] = \sum_{r=0}^{b-1} r\, \mathbb{P}(D_j^{(b)} = r).$$

The uniform term contributes

$$\frac{1}{b} \sum_{r=0}^{b-1} r = \frac{b-1}{2}.$$

For the first correction, use

$$\begin{aligned} \sum_{r=0}^{b-1} r(b-1-2r) &= (b-1) \sum_{r=0}^{b-1} r - 2 \sum_{r=0}^{b-1} r^2 \\ &= -\frac{b(b-1)(b+1)}{6}. \end{aligned} \tag{59}$$

Multiplying by the coefficient

$$\frac{b-1}{2b \ln b}\, b^{-j}$$

from (55) gives

$$-\frac{(b-1)^2(b+1)}{12 \ln b}\, b^{-j}.$$

The finite sum over r preserves the $O_b(b^{-2j})$ remainder. □

In particular,

$$\frac{b-1}{2} - \mathbb{E}[D_j^{(b)}] = O_b(b^{-j}), \tag{60}$$

and therefore

$$\sum_{j=1}^{\infty} \left| \frac{b-1}{2} - \mathbb{E}[D_j^{(b)}] \right| < \infty.$$

This completes the absolute-convergence assertion in Theorem 1.2.

4.3 A two-term expansion for the cumulative tail

We now return to the finite cumulative bias. The exact formula (44) permits substantially sharper information than the estimate

$$c_{b,L} = c_b + O_b(b^{-L})$$

used in Section 3.4.

For $x > 0$, write Stirling's expansion in the form

$$\begin{aligned} \ln \Gamma(x) = \left(x - \frac{1}{2}\right) \ln x - x + \frac{1}{2} \ln(2\pi) \\ + \frac{1}{12x} - \frac{1}{360x^3} + \rho(x). \end{aligned} \tag{61}$$

For positive real x, the standard first-neglected-term remainder bound for Stirling's series gives

$$0 < \rho(x) \leq \frac{1}{1260x^5}. \tag{62}$$

We use the positive-real remainder theorem in [9, §5.11(i)–(ii)].

Define

$$R(x) = \frac{1}{12x} - \frac{1}{360x^3} + \rho(x). \tag{63}$$

Proof of Theorem 1.3. Put

$$x = b^{L-1}.$$

Then

$$bx = b^L.$$

Substituting the Stirling expansion (61) into the exact formula (44), and subtracting from the limiting value (49), the elementary terms cancel exactly and give

$$c_b - c_{b,L} = \frac{bR(x) - R(bx)}{\ln b}. \tag{64}$$

Using (63),

$$\begin{aligned} bR(x) - R(bx) = \frac{b - b^{-1}}{12x} - \frac{b - b^{-3}}{360x^3} \\ + b\rho(x) - \rho(bx). \end{aligned} \tag{65}$$

Since

$$x = b^{L-1},$$

we have

$$\frac{b - b^{-1}}{x} = (b^2 - 1)b^{-L} \tag{66}$$

and

$$\frac{b - b^{-3}}{x^3} = (b^4 - 1)b^{-3L}. \tag{67}$$

Thus (64) becomes (7), with

$$\varepsilon_{b,L} = \frac{b\rho(x) - \rho(bx)}{\ln b}. \tag{68}$$

Because $b \geq 2$, one has $\ln b > 0$. Using (62) and the triangle inequality,

$$\begin{aligned} |\varepsilon_{b,L}| &\leq \frac{b\rho(x) + \rho(bx)}{\ln b} \\ &\leq \frac{1}{1260 \ln b} \left(\frac{b}{x^5} + \frac{1}{b^5 x^5} \right). \end{aligned} \tag{69}$$

Finally,

$$x = b^{L-1}$$

implies

$$\frac{b}{x^5} = b^6 b^{-5L}, \qquad \frac{1}{b^5 x^5} = b^{-5L},$$

and hence

$$|\varepsilon_{b,L}| \leq \frac{b^6 + 1}{1260 \ln b}\, b^{-5L}.$$

□

Remark. The absence of b^{-2L} and b^{-4L} terms in (7) reflects the odd inverse powers in the Stirling series for $\ln \Gamma(x)$.

4.4 Finite-prefix average

The preceding theorem sharpens (52) immediately.

Corollary 4.3. *For every $b \geq 2$ and $L \geq 1$,*

$$\boxed{\begin{aligned} \mathbb{E}[A_L^{(b)}] = \frac{b-1}{2} - \frac{c_b}{L} \\ + \frac{b^2 - 1}{12 \ln b} \frac{b^{-L}}{L} - \frac{b^4 - 1}{360 \ln b} \frac{b^{-3L}}{L} + \frac{\varepsilon_{b,L}}{L}, \end{aligned}} \tag{70}$$

where

$$|\varepsilon_{b,L}| \leq \frac{b^6 + 1}{1260 \ln b}\, b^{-5L}.$$

Proof. From (51),

$$\mathbb{E}[A_L^{(b)}] = \frac{b-1}{2} - \frac{c_{b,L}}{L}.$$

Write

$$c_{b,L} = c_b - (c_b - c_{b,L})$$

and apply Theorem 1.3. □

The expansion (70) shows that the L^{-1} correction determined by c_b is separated from exponentially small finite-prefix effects. This distinction will be useful in the deterministic setting: the next section asks whether the same $O(1)$ cumulative correction survives when the logarithmic mantissa integral is replaced by an orbit average along $\{n \log_b a\}$.

5 Deterministic exponential sequences and discrepancy transfer

We now pass from the logarithmic mantissa probability model to actual deterministic exponential sequences.

Let $a, b \geq 2$ be integers and assume that they are multiplicatively independent. Equivalently,

$$\alpha = \log_b a$$

is irrational. Define

$$\theta_n = \{n\alpha\}, \qquad n \geq 1. \tag{71}$$

For $\theta \in [0, 1)$, we write $D_j^{(b)}(b^\theta)$ for the j-th significant base-b digit of b^θ. Writing

$$n\alpha = \lfloor n\alpha \rfloor + \theta_n,$$

we have

$$a^n = b^{\lfloor n\alpha \rfloor} b^{\theta_n}. \tag{72}$$

Thus the significant digits of a^n are determined by the orbit (θ_n).

The number of base-b digits of a^n is

$$L_n = \lfloor n\alpha \rfloor + 1. \tag{73}$$

Throughout this section, $D_N(\alpha)$ denotes the ordinary interval discrepancy of the first N points of the rotation:

$$\boxed{D_N(\alpha) = \sup_{0 \leq u < v \leq 1} \left| \frac{1}{N} \#\{1 \leq n \leq N : \theta_n \in [u, v)\} - (v - u) \right|.} \tag{74}$$

We use ordinary discrepancy rather than star discrepancy because the digit events below are naturally unions of arbitrary half-open subintervals of $[0, 1)$.

5.1 Significant-digit cylinders

We first describe the interval geometry of a fixed significant-digit position.

For $j \geq 2$, put

$$M = b^{j-2}.$$

Fix

$$r \in \{0, \ldots, b - 1\}.$$

The condition that the j-th significant digit of b^θ is r is equivalent to the j-digit integer prefix having the form

$$bq + r, \qquad M \leq q \leq bM - 1.$$

For each such q, define

$$I_{q,r}^{(j)} = \left[\log_b \left(\frac{bq + r}{b^{j-1}} \right), \log_b \left(\frac{bq + r + 1}{b^{j-1}} \right) \right). \tag{75}$$

Lemma 5.1 (Digit-cylinder decomposition)**.** *For $j \geq 2$ and $r \in \{0, \ldots, b-1\}$, the event*

$$D_j^{(b)}(b^\theta) = r$$

is the disjoint union

$$\bigcup_{q=b^{j-2}}^{b^{j-1}-1} I_{q,r}^{(j)}. \tag{76}$$

In particular, it consists of exactly

$$\boxed{K_j = (b-1)b^{j-2}} \tag{77}$$

half-open intervals.

Proof. The first j significant base-b digits of b^θ form the integer

$$\left\lfloor b^{j-1+\theta} \right\rfloor.$$

Its final digit is r if and only if it equals $bq + r$ for some

$$b^{j-2} \leq q \leq b^{j-1} - 1.$$

For a fixed q,

$$\left\lfloor b^{j-1+\theta} \right\rfloor = bq + r$$

is equivalent to

$$bq + r \leq b^{j-1+\theta} < bq + r + 1,$$

which is exactly

$$\theta \in I_{q,r}^{(j)}.$$

Distinct prefix integers give disjoint half-open intervals. The number of admissible q is

$$b^{j-1} - b^{j-2} = (b-1)b^{j-2}.$$

□

The total length of the union in (76) is

$$\mathbb{P}(D_j^{(b)} = r),$$

the logarithmic-mantissa probability from Section 2.2.

For the leading digit, the corresponding cylinders are simply

$$I_d^{(1)} = [\log_b d, \log_b(d+1)), \qquad d = 1, \ldots, b-1. \tag{78}$$

5.2 Fixed-position discrepancy transfer

Lemma 5.1 gives an immediate quantitative comparison between the deterministic orbit and the logarithmic mantissa model.

Proposition 5.2 (Fixed-position transfer)**.** *For $j \geq 2$ and $r \in \{0, \dots, b-1\}$,*

$$\boxed{\left| \frac{1}{N} \#\{1 \leq n \leq N : D_j^{(b)}(b^{\theta_n}) = r\} - \mathbb{P}(D_j^{(b)} = r) \right| \leq (b-1) b^{j-2} D_N(\alpha).} \tag{79}$$

Proof. The digit event is the disjoint union of $K_j = (b-1)b^{j-2}$ intervals from Lemma 5.1. For each interval I, the definition of ordinary discrepancy gives

$$\left| \frac{1}{N} \#\{1 \leq n \leq N : \theta_n \in I\} - |I| \right| \leq D_N(\alpha).$$

Summing over the K_j disjoint components gives (79). □

To translate this into a statement about the actual integer a^n, we must account for the finitely many early exponents for which the j-th significant digit does not yet exist.

Define

$$e_j(N) = \#\{1 \leq n \leq N : L_n < j\}. \tag{80}$$

From

$$L_n < j \iff n\alpha < j - 1,$$

one has the exact formula

$$e_j(N) = \min\left\{ N,\, \max\left(0,\, \left\lceil \frac{j-1}{\alpha} \right\rceil - 1 \right) \right\}. \tag{81}$$

In particular,

$$e_j(N) = O_{a,b}(j). \tag{82}$$

Corollary 5.3. *For $j \geq 2$,*

$$\begin{aligned} &\left| \frac{1}{N} \#\{1 \leq n \leq N : L_n \geq j,\, D_j^{(b)}(a^n) = r\} - \mathbb{P}(D_j^{(b)} = r) \right| \\ &\qquad \leq (b-1) b^{j-2} D_N(\alpha) + \frac{e_j(N)}{N}. \end{aligned} \tag{83}$$

Proof. Whenever $L_n \geq j$, the first j significant digits of a^n are the first j significant digits determined by b^{θ_n}. Starting from Proposition 5.2, deleting the $e_j(N)$ short powers changes the normalized count by at most $e_j(N)/N$. □

Combining Corollary 5.3 with Proposition 4.1 gives, for fixed b,

$$\begin{aligned} &\frac{1}{N} \#\{1 \leq n \leq N : L_n \geq j,\, D_j^{(b)}(a^n) = r\} \\ &\quad = \frac{1}{b} + \frac{(b-1)(b-1-2r)}{2b \ln b}\, b^{-j} + O_b\big(b^{-2j} + b^j D_N(\alpha)\big) + O_{a,b}\left(\frac{j}{N}\right). \end{aligned} \tag{84}$$

This estimate is useful for orientation, but the main result below will sum the first J positions simultaneously.

5.3 The leading-window observable

For $J \geq 1$, define

$$G_J(\theta) = s_b\left(\left\lfloor b^{J-1+\theta} \right\rfloor\right), \qquad 0 \leq \theta < 1. \tag{85}$$

This is the sum of the first J significant digits of b^θ.

Lemma 5.4 (Exact leading-window coding)**.** *If $L_n \geq J$, then*

$$\boxed{G_J(\theta_n) = \sum_{j=1}^{J} D_j^{(b)}(a^n).} \tag{86}$$

Proof. From (72),

$$a^n = b^{L_n - 1} b^{\theta_n}.$$

Multiplying the normalized significand b^{θ_n} by b^{J-1} and taking the integer part extracts exactly the first J significant digits, provided those digits exist. Thus

$$\left\lfloor b^{J-1+\theta_n} \right\rfloor$$

is the J-digit integer prefix of a^n, and its digit sum is the right-hand side of (86). □

The logarithmic-mantissa average of G_J is already known exactly from Section 3.

Lemma 5.5 (Integral of the leading-window observable)**.** *For every $J \geq 1$,*

$$\boxed{\int_0^1 G_J(\theta)\, d\theta = J\frac{b-1}{2} - c_{b,J}.} \tag{87}$$

Proof. The random integer

$$\left\lfloor b^{J-1+\Theta} \right\rfloor$$

is precisely M_J from Section 2.1. Hence

$$\int_0^1 G_J(\theta)\, d\theta = \mathbb{E}[s_b(M_J)].$$

By the definition of $c_{b,J}$,

$$\mathbb{E}[s_b(M_J)] = J\frac{b-1}{2} - c_{b,J}.$$

□

5.4 A finite discrepancy bound for a leading window

Define the deterministic leading-window average

$$\mathcal{S}_{N,J}(a;b) = \frac{1}{N} \sum_{\substack{1 \leq n \leq N \\ L_n \geq J}} \sum_{j=1}^{J} D_j^{(b)}(a^n). \tag{88}$$

Before deleting the short powers, consider

$$T_{N,J} = \frac{1}{N} \sum_{n=1}^{N} G_J(\theta_n).$$

Since

$$G_J(\theta) = \sum_{j=1}^{J} D_j^{(b)}(b^\theta),$$

we may estimate each digit position by its interval cylinders.

For the first digit, (78) and ordinary discrepancy give

$$\begin{aligned} \left| \frac{1}{N} \sum_{n=1}^{N} D_1^{(b)}(b^{\theta_n}) - \mathbb{E}[D_1^{(b)}] \right| &\le \sum_{d=1}^{b-1} d\, D_N(\alpha) \\ &= \frac{b(b-1)}{2} D_N(\alpha). \end{aligned} \tag{89}$$

For $j \ge 2$, there are

$$K_j = (b-1)b^{j-2}$$

components for each digit r. Hence

$$\begin{aligned} &\left| \frac{1}{N} \sum_{n=1}^{N} D_j^{(b)}(b^{\theta_n}) - \mathbb{E}[D_j^{(b)}] \right| \\ &\quad \le K_j D_N(\alpha) \sum_{r=0}^{b-1} r \\ &\quad = \frac{b(b-1)}{2}(b-1)b^{j-2} D_N(\alpha). \end{aligned} \tag{90}$$

Summing (89) and (90) over $1 \le j \le J$, the geometric series collapses:

$$\begin{aligned} &\frac{b(b-1)}{2} + \sum_{j=2}^{J} \frac{b(b-1)}{2}(b-1)b^{j-2} \\ &\quad = \frac{b(b-1)}{2} \left(1 + (b-1) \sum_{j=2}^{J} b^{j-2} \right) \\ &\quad = \frac{b-1}{2} b^J. \end{aligned} \tag{91}$$

Therefore

$$\left| T_{N,J} - \int_0^1 G_J(\theta)\, d\theta \right| \le \frac{b-1}{2} b^J D_N(\alpha). \tag{92}$$

We can now prove the finite deterministic transfer theorem announced in the introduction.

Proof of Theorem 1.4. By Lemma 5.4,

$$\mathcal{S}_{N,J}(a;b)$$

agrees with $T_{N,J}$ on every exponent satisfying $L_n \ge J$. The two quantities differ only at the

$$e_J(N)$$

short powers.

Since

$$0 \le G_J(\theta) \le J(b-1),$$

we have

$$|\mathcal{S}_{N,J}(a;b) - T_{N,J}| \le \frac{(b-1)J\,e_J(N)}{N}. \tag{93}$$

Combining (92), (93), and Lemma 5.5 gives

$$\boxed{\left|\mathcal{S}_{N,J}(a;b) - \left(J\frac{b-1}{2} - c_{b,J}\right)\right| \le \frac{b-1}{2}b^J D_N(\alpha) + \frac{(b-1)J\,e_J(N)}{N}.}$$

□

5.5 The second-order deterministic term

The finite bound immediately isolates the cumulative Benford bias as the bounded correction term.

Proof of Theorem 1.5. From Theorem 1.4,

$$\mathcal{S}_{N,J}(a;b) = J\frac{b-1}{2} - c_{b,J} + O_b\big(b^J D_N(\alpha)\big) + O_{a,b}\left(\frac{J^2}{N}\right),$$

where we used

$$e_J(N) = O_{a,b}(J).$$

By Theorem 1.3,

$$c_{b,J} = c_b + O_b(b^{-J}).$$

Consequently,

$$\boxed{\mathcal{S}_{N,J}(a;b) = J\frac{b-1}{2} - c_b + O_b(b^{-J}) + O_b\big(b^J D_N(\alpha)\big) + O_{a,b}\left(\frac{J^2}{N}\right).} \tag{94}$$

If

$$J = J(N) \longrightarrow \infty,$$
$$b^{J(N)} D_N(\alpha) \longrightarrow 0,$$

and

$$\frac{J(N)^2}{N} \longrightarrow 0,$$

then every error term in (94) tends to zero. Hence

$$\boxed{\mathcal{S}_{N,J(N)}(a;b) = J(N)\frac{b-1}{2} - c_b + o(1).}$$

□

Equivalently,

$$\boxed{J(N)\frac{b-1}{2} - \mathcal{S}_{N,J(N)}(a;b) \longrightarrow c_b.} \tag{95}$$

Thus the same constant obtained by integrating against the logarithmic mantissa measure survives as the bounded correction in a deterministic orbit average.

5.6 A power-saving discrepancy input

The preceding theorem is formulated conditionally in terms of $D_N(\alpha)$. For a fixed multiplicatively independent pair (a, b), classical Diophantine-discrepancy theory supplies the required quantitative input.

Fuchs, building on a result of Kiss and Tichy and Baker's theory of linear forms in logarithms, records the following power-saving discrepancy estimate for irrational quotients of logarithms [12, Lemma 2].

Proposition 5.6 (Classical power-saving discrepancy)**.** *Let $a, b \geq 2$ be fixed multiplicatively independent integers. Then there exists a constant*

$$\gamma_{a,b} > 0$$

such that

$$\boxed{D_N(\log_b a) \ll_{a,b} N^{-\gamma_{a,b}}.} \tag{96}$$

Remark (External input). Proposition 5.6 is not a result of this paper. Fuchs states the estimate for

$$\alpha = \log_A Q$$

and the sequence $-n\alpha$, as a special case of a more general result of Kiss and Tichy, and notes that the proof follows from the Erdős–Turán inequality together with Baker's theorem on linear forms in logarithms [12, 16, 17]. Taking

$$A = b, \qquad Q = a$$

gives $\alpha = \log_b a$. The sign is immaterial for ordinary discrepancy, since reflection modulo one maps intervals to intervals of the same length.

5.7 Logarithmically growing leading windows

We can now remove the discrepancy hypothesis from the main second-order statement for each fixed pair (a, b).

Proof of Corollary 1.6. Fix

$$0 < \kappa < \gamma_{a,b}$$

and define

$$J_N = \lfloor \kappa \log_b N \rfloor .$$

Then

$$J_N \longrightarrow \infty$$

and

$$\frac{J_N^2}{N} = O\left(\frac{(\log N)^2}{N} \right) \longrightarrow 0.$$

Moreover,

$$b^{J_N} \leq N^{\kappa}.$$

Using Proposition 5.6,

$$b^{J_N} D_N(\log_b a) \ll_{a,b} N^{\kappa - \gamma_{a,b}} \longrightarrow 0.$$

The hypotheses of Theorem 1.5 are therefore satisfied, and

$$\boxed{\mathcal{S}_{N,J_N}(a;b) = J_N \frac{b-1}{2} - c_b + o(1).}$$

□

The proof also gives an explicit qualitative rate.

Corollary 5.7 (Quantitative logarithmic-window form)**.** *Under the hypotheses above,*

$$\mathcal{S}_{N,J_N}(a;b) = J_N \frac{b-1}{2} - c_b + O_{a,b,\kappa}\left(N^{-\kappa} + N^{-(\gamma_{a,b}-\kappa)} + \frac{(\log N)^2}{N}\right). \tag{97}$$

Proof. Since

$$J_N \geq \kappa \log_b N - 1,$$

we have

$$b^{-J_N} \ll_b N^{-\kappa}.$$

The other two terms follow directly from (94) and (96). □

For example, choosing

$$\kappa = \frac{\gamma_{a,b}}{2}$$

balances the first two power-saving errors and yields

$$\mathcal{S}_{N,J_N}(a;b) = J_N \frac{b-1}{2} - c_b + O_{a,b}\left(N^{-\gamma_{a,b}/2} + \frac{(\log N)^2}{N}\right), \tag{98}$$

with

$$J_N = \left\lfloor \frac{\gamma_{a,b}}{2} \log_b N \right\rfloor.$$

5.8 Relation to earlier growing-window results

The logarithmic growth of the controlled window is not itself new. Fuchs studied digital blocks in growing regions of exponential sequences and used discrepancy and exponential-sum estimates to obtain first-order asymptotic digit statistics [12]. The discrepancy machinery in Proposition 5.6 likewise belongs to that earlier literature.

The point of (95) is instead second-order. The first-order term

$$J\frac{b-1}{2}$$

reflects the asymptotically uniform bulk digit distribution. After this term is removed, the remaining bounded contribution is not discarded inside an $O(1)$ error: it converges to

$$-c_b.$$

Thus

$$c_b = \frac{b}{2} - \frac{b-1}{2}\log_b(2\pi)$$

acts as a boundary correction generated by the finitely biased significant positions near the leading edge.

To the best of our knowledge, this explicit constant term has not previously been recorded in the growing-window theory for exponential sequences.

5.9 Scope of the deterministic result

It is important to distinguish the leading-window theorem from a statement about the complete base-b digit sum of a^n.

For n of order N,

$$L_n = n \log_b a + O(1) \asymp_{a,b} N,$$

whereas the unconditional window obtained above has only

$$J_N = O_{a,b}(\log N)$$

positions. Hence

$$\frac{J_N}{L_n} \longrightarrow 0$$

for typical n in the averaging range.

Accordingly, the present argument controls a vanishing fraction of the complete digit expansion. It does not establish a mean law, variance law, central limit theorem, mixing property, or independence statement for the full digit sum $s_b(a^n)$. Extending absolute-error control from a logarithmic leading window to a positive proportion of the digits would require substantially stronger input than the discrepancy estimate used here.

6 Computational verification and reproducibility

The results of Sections 3–5 are proved analytically and do not depend on numerical computation. Nevertheless, because several analytic identities involve substantial cancellation and the deterministic estimates require delicate indexing, boundary handling, and discrepancy bookkeeping, we carried out a separate adversarial computational audit of the full theorem package.

The purpose of this audit was not to supply numerical evidence in place of proof. Rather, it was designed to detect indexing errors, normalization mistakes, hidden boundary cases, numerical instability, or disagreements between mathematically equivalent formulations.

6.1 Audit structure

The verification package treats the principal statements as sixteen separate claims, denoted internally by

$$\mathrm{A1}, \ldots, \mathrm{A10}, \qquad \mathrm{D1}, \ldots, \mathrm{D6}.$$

The analytic claims include the exact significant-digit formulas, the finite Benford-weighted digit-sum identity, the closed expression for c_b, the deep-position expansions, and the finite-length Stirling remainder estimate. The deterministic claims include the digit-cylinder decomposition, leading-window coding, and the finite discrepancy transfer estimates.

Where possible, the same mathematical quantity was recomputed through independent representations. In particular, the audit compared:

- direct logarithmic interval sums with Gamma-function formulas;
- position-wise digit expectations with direct enumeration of finite integer significands;
- the finite cumulative bias obtained from individual digits with the factorial identity of Theorem 1.1;

- mantissa-based digit extraction with exact integer base-b digit extraction for powers a^n;
- the theoretical discrepancy bounds with explicit orbit counts for finite test cases.

All comparisons involving Gamma functions or deep Stirling tails were performed at arbitrary precision rather than ordinary machine floating-point precision.

6.2 Verification results

No candidate identity in the final theorem package was falsified. All sixteen claims passed the final audit.

For the analytic part, the finite prefix probabilities, Gamma identities, factorial identity, and closed formula for c_b agreed across independent computational routes to high precision. The deep-position coefficients in Proposition 4.1 and Corollary 4.2 were also checked by scaled residual experiments across several bases.

The finite-tail theorem Theorem 1.3 received a separate high-precision stress test. Because the relevant expression involves subtraction of large log-Gamma terms, low working precision can produce spurious violations of the remainder estimate. The final experiments therefore used 220 decimal digits of precision. The numerical tests were consistent with the rigorous bound

$$|\varepsilon_{b,L}| \le \frac{b^6+1}{1260\ln b}\, b^{-5L}.$$

These experiments are supplementary only: the proof of the bound is the analytic positive-real Stirling remainder argument in Section 4.3.

For the deterministic part, exact integer digit extraction was compared against the mantissa representation for thousands of powers and their tested leading positions, with no disagreement after explicit guarding of exact and near-exact integer boundaries. The coefficient

$$\frac{b-1}{2}b^J$$

in the finite leading-window discrepancy bound was also independently reconstructed from the weighted interval-component counts and tested against direct finite orbit computations.

The final automated test suite contained sixteen tests, all of which passed.

6.3 Numerical boundary issues

Two numerical issues encountered during the audit are worth recording.

First, evaluating

$$\left\lfloor b^{J-1+\theta} \right\rfloor$$

directly with finite-precision floating-point arithmetic can be unreliable when the exact value is an integer or lies extremely close to one. Exact integer arithmetic was therefore treated as the ground truth for digit extraction, and high-precision nearest-integer checks were used at mantissa boundaries.

Second, direct evaluation of the finite-bias formula

$$\ln\Gamma(b^L) - b\ln\Gamma(b^{L-1})$$

can suffer severe cancellation for large L. High-precision arithmetic was used in all tests involving the exponentially small tail $c_b - c_{b,L}$.

These observations concern implementation only and do not alter the analytic statements proved above.

6.4 Reproducibility archive

The computational audit is organized as a separate reproducibility repository containing:

- an authoritative theorem specification matching the claims proved in the manuscript;
- independent implementations of the principal formulas;
- automated unit and adversarial tests;
- high-precision residual experiments for the asymptotic statements;
- deterministic digit-extraction and discrepancy experiments;
- machine-readable claim-status and numerical-residual tables;
- instructions sufficient to recreate the final verification run.

The verification code is intentionally separated from the proofs in this paper. Its role is to make the formulas independently checkable and to document numerical edge cases, while the mathematical validity of the main theorems rests entirely on the arguments in Sections 3–5.

7 Discussion, limitations, and open problems

The results of this paper identify a precise second-order contribution to the mean sum of a growing block of leading significant digits in an exponential sequence. The main mechanism can be summarized as follows: the early significant positions are not individually uniform under the logarithmic mantissa measure, their cumulative signed deviation has the finite limit

$$c_b = \frac{b}{2} - \frac{b-1}{2}\log_b(2\pi),$$

and quantitative equidistribution transfers this same bounded correction to a deterministic leading-window average.

This section records the limits of that argument and several natural directions in which the present results might be extended.

7.1 Leading windows versus complete digit sums

The principal deterministic result concerns

$$\mathcal{S}_{N,J}(a;b) = \frac{1}{N}\sum_{\substack{1\le n\le N\\ L_n\ge J}}\sum_{j=1}^{J} D_j^{(b)}(a^n),$$

with

$$J = J(N)$$

chosen so that

$$b^J D_N(\log_b a) \longrightarrow 0.$$

For the unconditional fixed-pair corollary obtained from a power-saving discrepancy estimate, this gives

$$J_N = O_{a,b}(\log N).$$

By contrast, for $n \asymp N$, the complete number of base-b digits of a^n is

$$L_n = n \log_b a + O(1) \asymp_{a,b} N.$$

Hence the controlled leading block satisfies

$$\frac{J_N}{L_n} \longrightarrow 0.$$

The theorem therefore does not determine the asymptotic behavior of the complete sum of digits

$$s_b(a^n) = \sum_{j=1}^{L_n} D_j^{(b)}(a^n),$$

nor of the normalized complete-digit average

$$\frac{s_b(a^n)}{L_n}.$$

Controlling a logarithmic number of significant positions is sufficient to retain the finite boundary constant c_b, but it controls only a vanishing fraction of the full expansion.

This distinction is essential: the second-order theorem proved here should not be interpreted as a complete digit-sum theorem for exponential sequences.

7.2 The barrier imposed by discrepancy

The finite deterministic estimate

$$\mathcal{S}_{N,J}(a;b) = J\frac{b-1}{2} - c_b + O_b(b^{-J}) + O_b\big(b^J D_N(\log_b a)\big) + O_{a,b}\left(\frac{J^2}{N}\right)$$

makes the present limitation explicit.

The complexity of the first J significant positions grows essentially like b^J. Ordinary discrepancy must therefore decrease fast enough to compensate for this exponential growth. A polynomial bound

$$D_N(\log_b a) \ll N^{-\gamma}$$

naturally leads to a logarithmic admissible range

$$J \lesssim \gamma \log_b N.$$

Thus one obvious open problem is to determine whether stronger arithmetic or harmonic analysis can control significantly larger leading windows while retaining an absolute $o(1)$ error after subtracting the first-order term.

The difficulty is stronger than proving first-order digit equidistribution. A statement of the form

$$\mathcal{S}_{N,J} = J\frac{b-1}{2} + o(J)$$

does not determine a bounded second-order term. To identify $-c_b$, the total error must instead be controlled on the absolute scale $o(1)$.

7.3 Beyond logarithmic windows

A natural first extension would be to replace

$$J_N \asymp \log N$$

by a faster-growing sequence while preserving

$$\mathcal{S}_{N,J_N}(a;b) = J_N \frac{b-1}{2} - c_b + o(1).$$

There are at least two conceptually different ways such an improvement could arise. One could seek sharper discrepancy estimates tailored to the specific rotation

$$\{n \log_b a\},$$

or one could exploit more structure in the digit observable itself, rather than bounding all of its interval components independently.

The present proof deliberately uses ordinary discrepancy in a robust but coarse way. It does not exploit possible cancellation between the many digit cylinders. Whether such cancellation can enlarge the second-order window remains open.

7.4 The complete average-digit problem

A substantially harder problem is to understand

$$A_{a,b}(n) = \frac{s_b(a^n)}{L_n}. \tag{99}$$

The logarithmic mantissa model suggests the uniform digit mean

$$\frac{b-1}{2}$$

as the natural center, but the present results do not prove

$$A_{a,b}(n) \to \frac{b-1}{2}$$

pointwise, in mean over n, or in any stronger sense.

The obstacle is structural. For each n, the statistic (99) depends on a number of digit positions that grows linearly with n. Standard equidistribution of

$$\{n \log_b a\}$$

controls every fixed observable, while the complete digit average corresponds to a family of observables whose complexity increases with n.

Bridging this varying-observable gap appears to require ideas beyond the fixed-position and logarithmic-window methods used here.

7.5 Fluctuations and dependence

The present paper concerns expectations, cumulative bias, and deterministic orbit averages. It does not establish a variance law or a limiting distribution for digit sums of a^n.

In particular, no assumption of independence among significant digits is made. Irrational rotations are equidistributed but are not mixing, and fixed-position marginal equidistribution does not imply temporal independence, Gaussian fluctuations, or a central limit theorem for a changing digit observable.

Natural future questions include whether appropriately normalized leading-window or complete-digit statistics possess limiting variance laws, whether arithmetic properties of a and b influence those fluctuations, and whether any probabilistic limit theorem is available under additional hypotheses.

These questions are deliberately left separate from the deterministic second-order theory proved above.

7.6 Other digital functions

The finite identity in Theorem 1.1 arises from two ingredients: the logarithmic Benford weight

$$\log_b\left(\frac{m+1}{m}\right)$$

and the radix recursion

$$s_b(bq+r) = s_b(q) + r.$$

This suggests a broader question.

Let f be a digital or b-additive function. One may ask whether finite logarithmically weighted expectations

$$\sum_{m=b^{L-1}}^{b^L-1} f(m)\log_b\left(\frac{m+1}{m}\right)$$

admit exact product, Gamma-function, or asymptotic evaluations analogous to those obtained here for s_b.

Such extensions would connect the present Benford-weighted calculation more directly with the classical theory of digital and q-additive functions. We do not pursue them here.

7.7 Summary of the boundary correction

The principal phenomenon established in this paper is a summable boundary effect.

The significant positions closest to the leading edge retain a systematic logarithmic bias. Although the bias at position j decays exponentially with j, the sum of these deviations converges to the finite constant

$$c_b.$$

Consequently, the mean sum of the first J significant digits is not merely

$$J\frac{b-1}{2} + o(1),$$

but rather

$$J\frac{b-1}{2} - c_b + o(1)$$

whenever the underlying orbit is sampled with sufficient quantitative equidistribution.

This provides a precise sense in which the logarithmic structure of the leading digits survives beyond the first-order uniform-digit law. The extent to which analogous second-order effects persist deeper into the full digital expansion remains open.

8 Conclusion

This paper studied the cumulative effect of the nonuniform early significant digits under the logarithmic, or Benford, mantissa measure. Starting from the classical distribution of finite significant prefixes, we obtained an exact logarithmically weighted finite digit-sum identity and used it to evaluate the cumulative signed significant-digit bias in closed form:

$$\boxed{c_b = \frac{b}{2} - \frac{b-1}{2}\log_b(2\pi).}$$

The same finite identity also yields explicit control of the approach to this limit through the Stirling expansion and its positive-real remainder bound.

The second part of the paper transferred this model-side constant to deterministic exponential sequences. For multiplicatively independent integers $a, b \geq 2$, quantitative equidistribution of $\{n \log_b a\}$ allows a logarithmically growing block of leading significant digits to be averaged with absolute second-order accuracy. For an admissible growing window J_N, this gives

$$\boxed{\mathcal{S}_{N,J_N}(a;b) = J_N \frac{b-1}{2} - c_b + o(1).}$$

Thus the bounded correction predicted by the logarithmic mantissa model survives in an actual deterministic orbit.

The first-order ingredients behind this result belong to established theory: finite significant-digit laws, higher-digit Gamma representations, uniform distribution modulo one, and discrepancy methods for exponential sequences all have substantial prior literature. The contribution developed here is instead the finite Benford-weighted digit-sum evaluation, its closed cumulative constant, and the retention of that same constant as a second-order term in the controlled deterministic leading window.

The present method does not address the complete digit expansion of a^n, for which the number of digits grows linearly with n, nor does it establish variance laws, independence, mixing, or probabilistic limit theorems for full digit sums. Extending second-order control beyond logarithmically growing leading windows, and understanding whether related boundary corrections occur for more general digital functions, remain natural problems for future work.

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